



Missouri University of Science and Technology

Vector Similarity Search

Shape Search

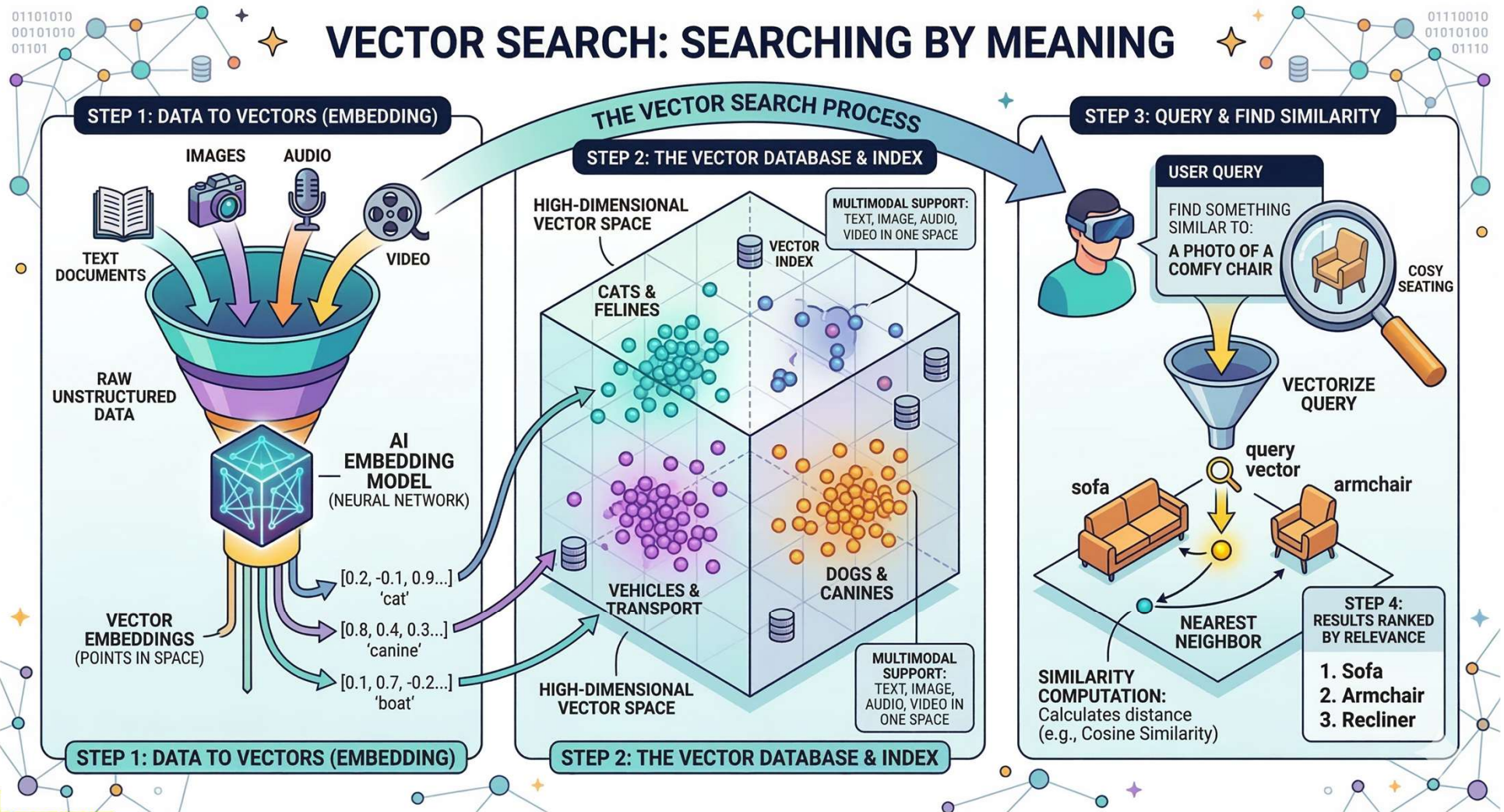
Jaccard Similarity

Vector Search

- 1) Text => near duplicate detection
- 2) Feature Vectors => Top-K search
- 3) Shape Similarity Search



VECTOR SEARCH: SEARCHING BY MEANING

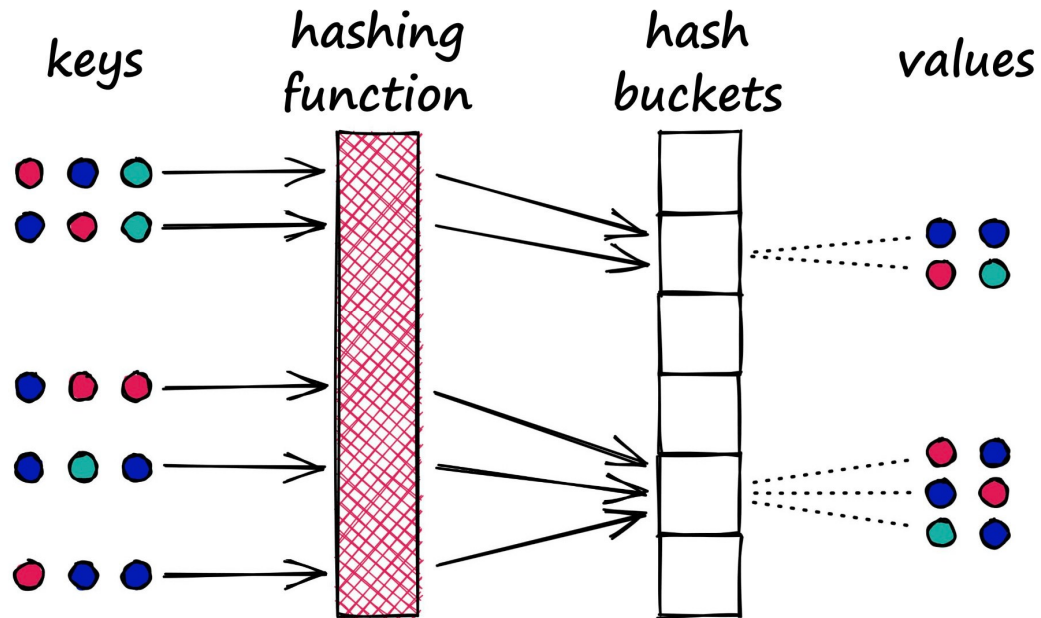


Block diagram Vector Search System

Raw Data → “Encoding” → Vectors → “MinHashing” → Short Signatures → “LSH” → Hash Table for search



Locality Sensitive Hashing



An LSH function aims to place similar values into the same buckets.

1.1 Locality Sensitive Functions

Let $d_1 < d_2$ be two distances according to Jaccard distance measure d . A family F of functions is said to be (d_1, d_2, p_1, p_2) -sensitive if for every f in F :

- if $d(p, q) \leq d_1$, then $\Pr(h(p) = h(q)) \geq p_1$
- if $d(p, q) \geq d_2$, then $\Pr(h(p) = h(q)) \leq p_2$

where $p_1 > p_2$.

Locality sensitive hashing

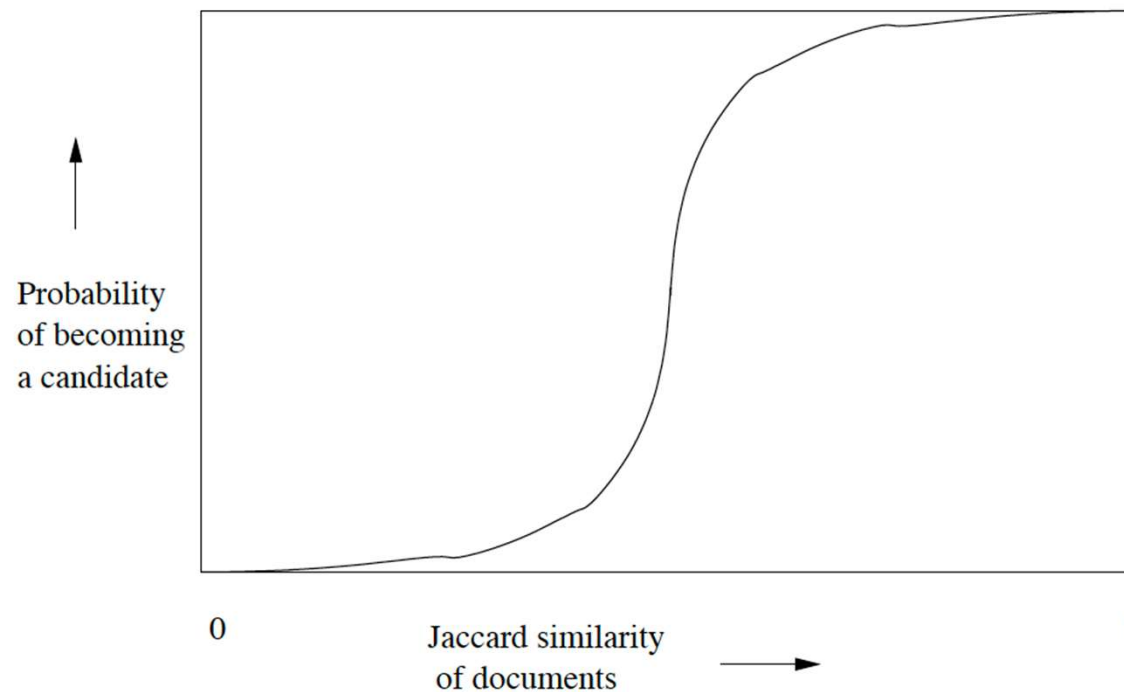


Figure 3.8: The S-curve

What is Jaccard Similarity

Set similarity $\Rightarrow$ set intersection/set union

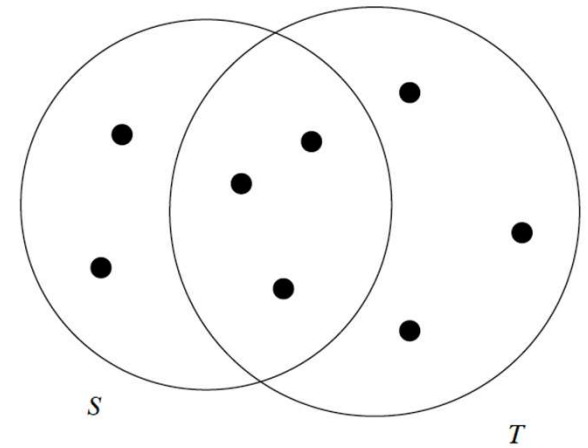


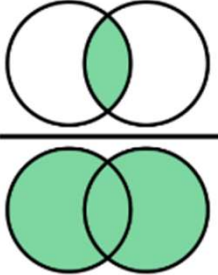
Figure 3.1: Two sets with Jaccard similarity 3/8

Vector similarity $\Rightarrow$ intersection/union

Shape similarity $\Rightarrow$ area of intersection/area of union

What is Jaccard Similarity?

Convert large sets to short signatures, while preserving Jaccard similarity

$$\text{Jaccard Similarity} = \frac{\text{Overlap}}{\text{Total Size}}$$


Jaccard similarity measures how similar two sets are by comparing their overlap relative to their total size.

Vector similarity via Jaccard

Jaccard Similarity between two vectors A and B

$$J_w(A, B) = \frac{\sum_{i=1}^n \min(A_i, B_i)}{\sum_{i=1}^n \max(A_i, B_i)}$$

What is MinHashing?

Signatures /MinHashes

Key idea: “hash” each set S to a small signature $\text{Sig}(S)$, such that:

1. $\text{Signature}(S)$ is small enough that we can fit a signature in main memory for each set.
2. $\text{Sim}(S1, S2)$ is (almost) the same as the “similarity” of $\text{Signature}(S1)$ and $\text{Signature}(S2)$ i.e minhash codes. (signature preserves similarity).

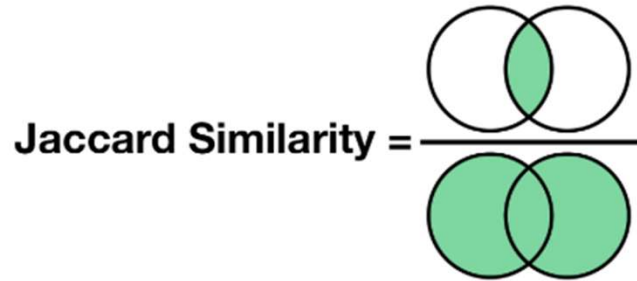
Shape Search

Shape

Area

Shape s_1 , Shape s_2 .

$$\text{Jaccard Similarity}(s_1, s_2) = \frac{\text{Area of Intersection}(s_1, s_2)}{\text{Area of Union}(s_1, s_2)}$$



Shape similarity search

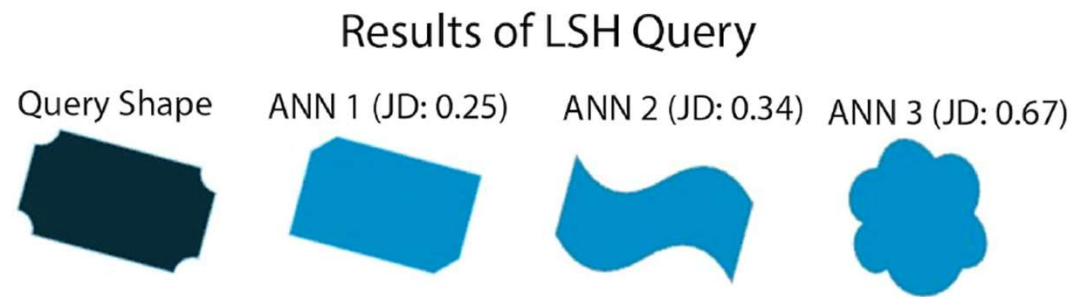


Fig. 1. An example of a K-Approximate nearest-neighbor (K-ANN) query and its results for K=3. JD is the Jaccard distance. In GIS, the geometries are represented as a list of coordinates (latitude, longitude).

Example of shape similarity search



Figure 1: Top- K Approximate Nearest Neighbor (ANN) query example using Land dataset. Results are sorted by distance for $K=3$. JD is the Jaccard distance ($1 - \text{Jaccard similarity}$).



PolyMinHash: Shape MinHashing for Approximate Nearest Neighbor Search-

Alima Subedi, Sankalpa Pokharel & Satish Puri

Missouri University of Science and Technology

Why Polygon Similarity Search?

Finding similar shape is a critical task in GIS. it's incredibly slow.

Problem

The brute-force method is unfeasibly slow because it requires expensive geometric calculations (intersection and union).

GAP

- MinHash is a tool for finding similar *text* or *images* , it just doesn't work out-of-the-box for polygon *areas*.
- Existing methods can't be used because polygons are closed shapes and require a concept of area.

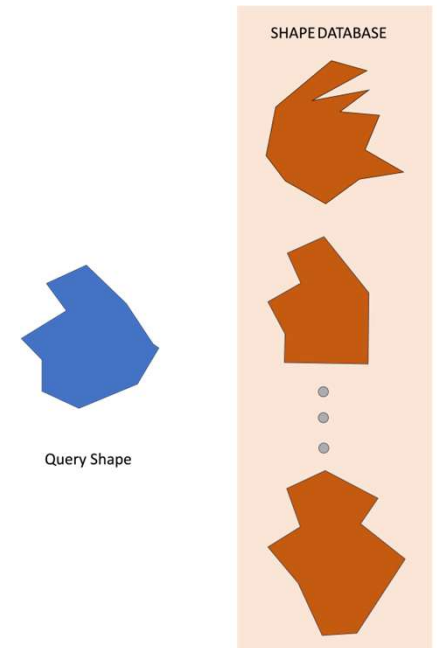


Figure 1: The Brute-Force (Exact) Search Problem

MinHash for Shapes

Collision Probability of two shapes $s1$ and $s2$ is given by

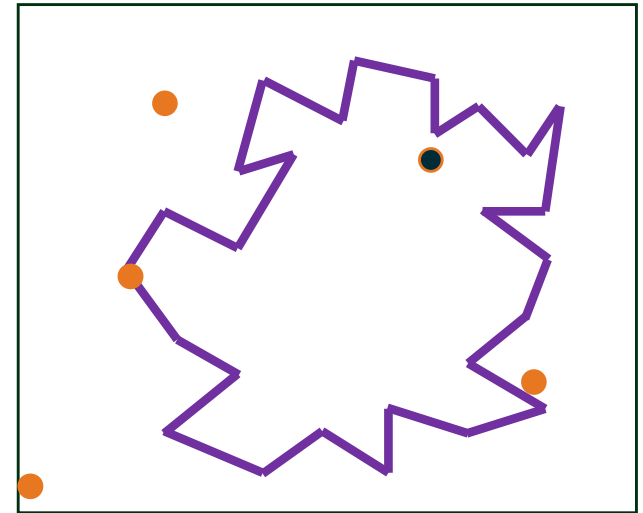
Area of Intersection($s1, s2$)

$\Pr(h(s1) = h(s2)) = \text{-----}$

Area of Union($s1, s2$)

How PolyMinHash Works

- Inspired by WMH, we **adapted the method** directly for 2D geometries.
- Core idea: **Random sampling + rejection** — but done **geometrically**.
- This lead to **Dart Sampling** for polygons.



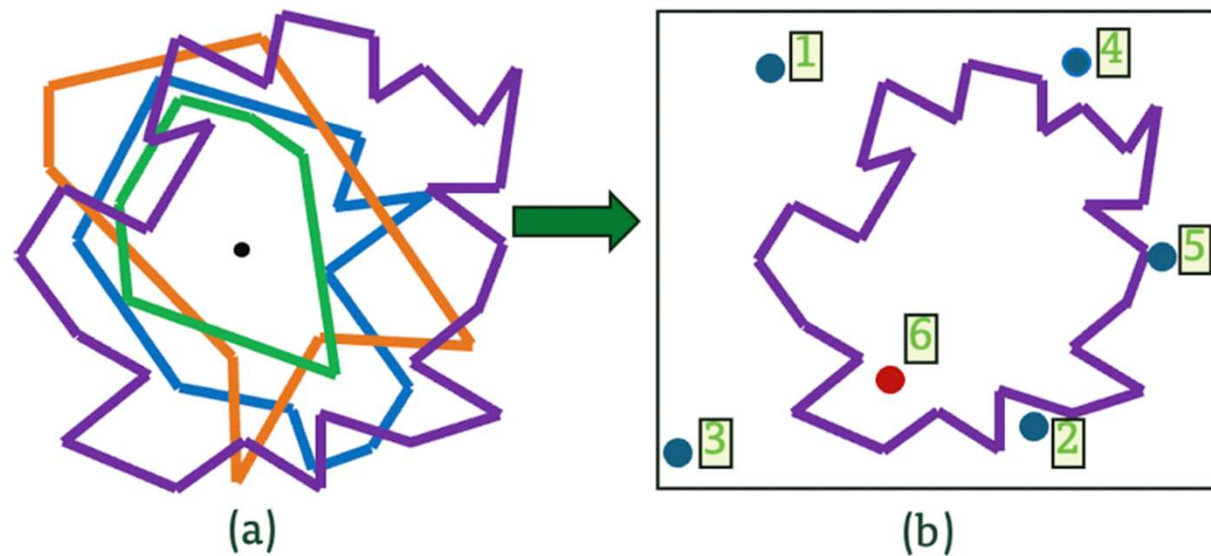
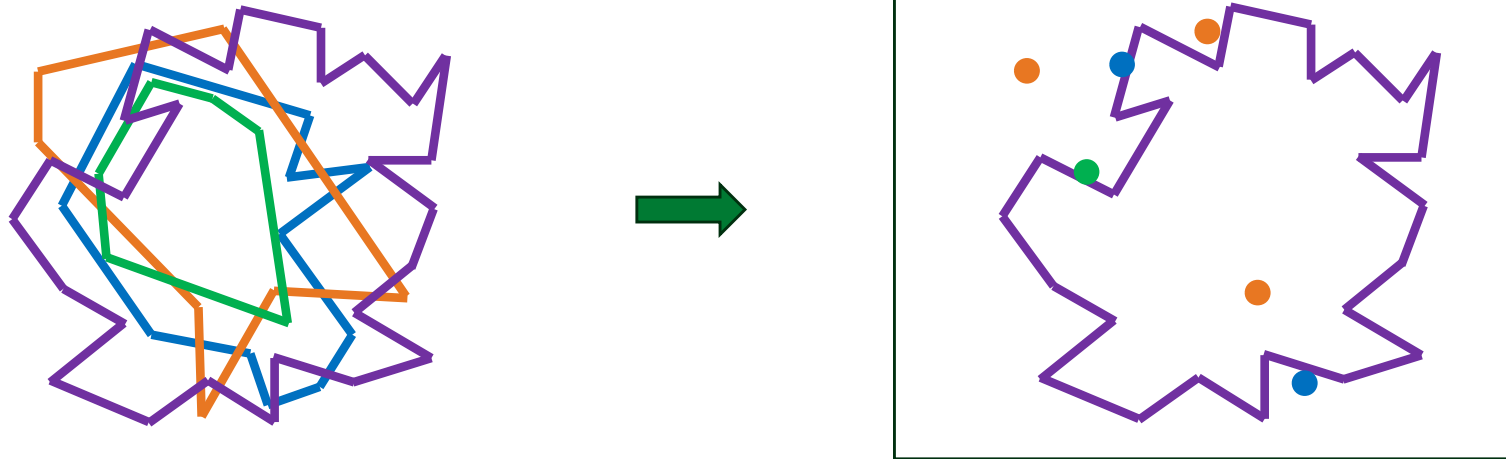


Fig. 3. Overview of PolyMinHash. (a) Center the polygons. (b) Construct the global MBR B and sample 2D points randomly within B for Polygon P . The red dot #6 is the first point that lands inside the polygon. Therefore, the $\text{MinHash}(P)$ is 6.

Our Method: PolyMinHash

- MinHashing method for 2d shapes.
- Core idea: **Unbiased random sampling + rejection** — but done **geometrically**



Global Sampling: Global Minimum Bounding Rectangle (MBR) acts like total domain.

MinHash Collision Probability

$$\begin{aligned}\Pr[h(P) = h(Q)] &= \frac{\text{Area}(P \cap Q)}{\text{Area}(P \cap Q) + \text{Area}(P \setminus Q) + \text{Area}(Q \setminus P)} \\ &= \frac{\text{Area}(P \cap Q)}{\text{Area}(P \cup Q)} = J(P, Q)\end{aligned}$$

Extension to Vector Search

Similarity Search

Find top-K vectors similar to Vector x .



Missouri University of Science and Technology

Histogram Similarity

Jaccard Similarity between two vectors A and B

$$J_w(A, B) = \frac{\sum_{i=1}^n \min(A_i, B_i)}{\sum_{i=1}^n \max(A_i, B_i)}$$

Jaccard distance (A, B) = 1 - Jw

Weighted MinHashing

Motivation

Standard MinHash works for **binary sets**

Many data types are **real-valued / weighted**

Key Property

$$\Pr[h(\mathbf{x}) = h(\mathbf{y})] = \frac{\sum_i \min(x_i, y_i)}{\sum_i \max(x_i, y_i)}$$

Collision probability equals **Weighted Jaccard similarity**

Simple and Weighted MinWise Hashing

1. Rejection Sampling
2. Randomly sample points over a "red-green" map of the vector.
3. Count how many tries it takes to hit the green (nonzero) region — that count becomes the hash value.

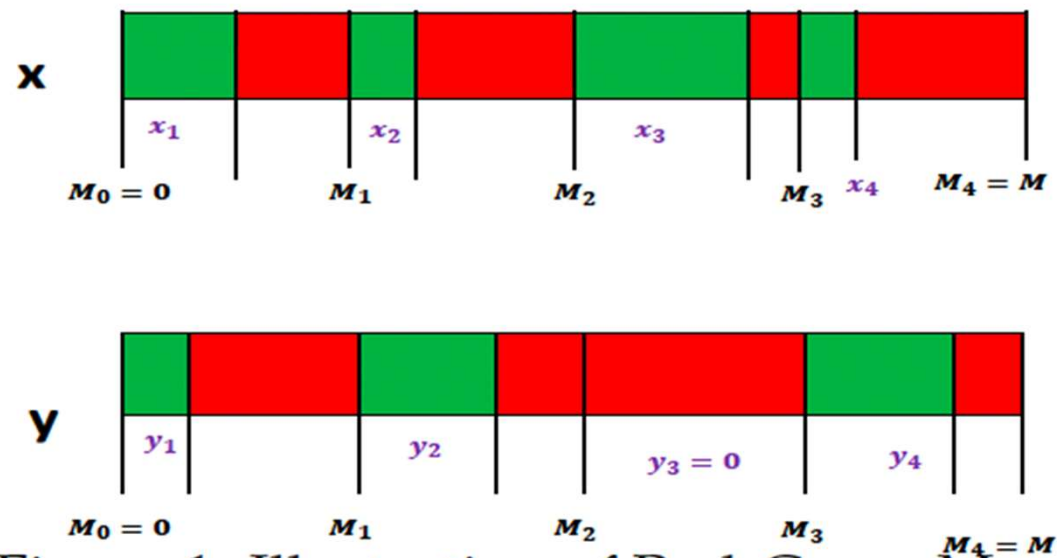
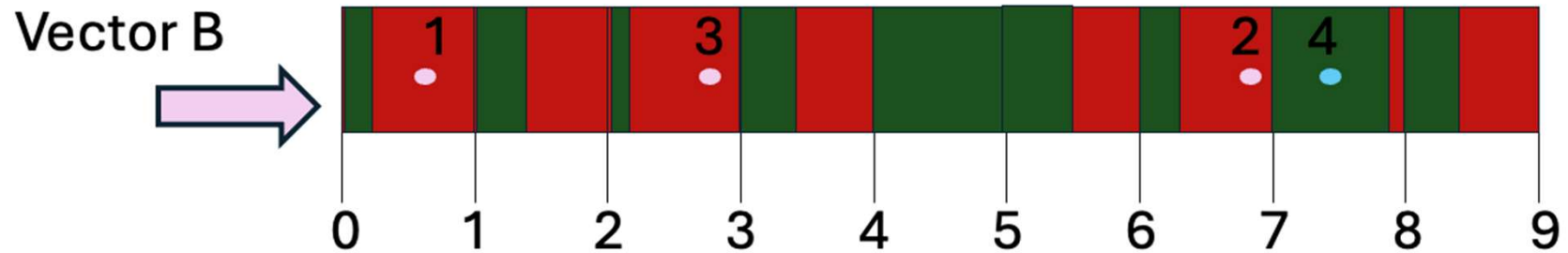
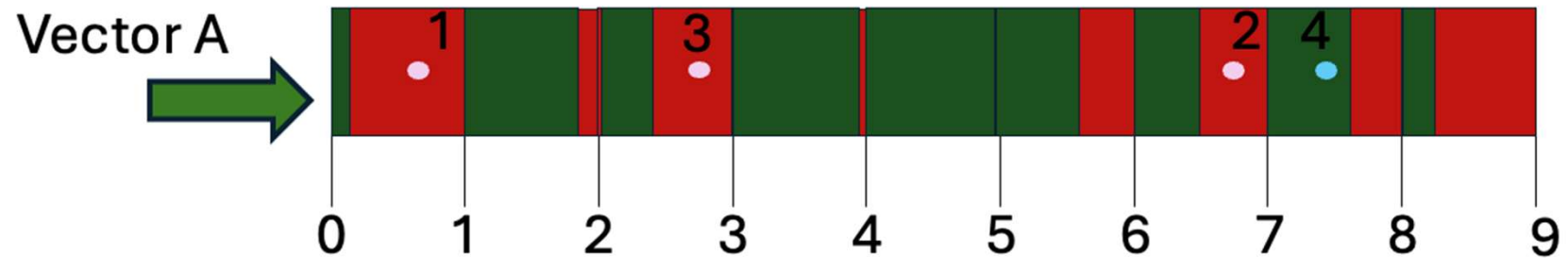
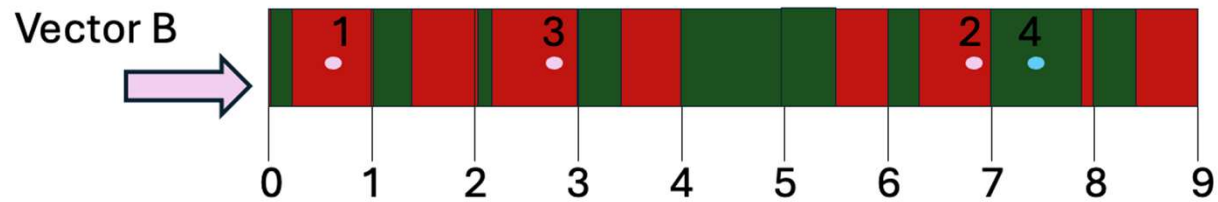
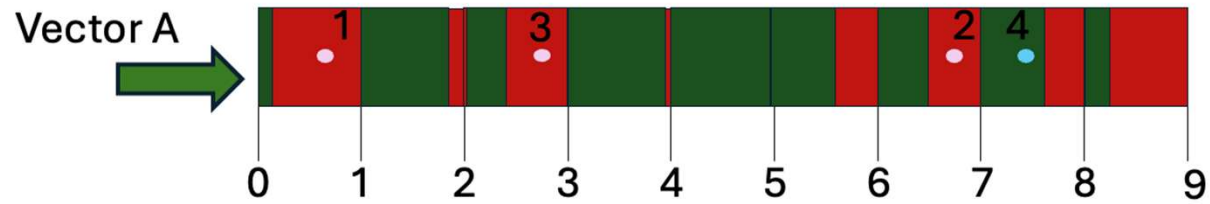


Figure 1: Illustration of Red-Green Map of 4 dimensional vectors x and y .

Illustration of Rejection Sampling for Vectors





$$J_w(A, B) = \frac{\sum_{i=1}^n \min(A_i, B_i)}{\sum_{i=1}^n \max(A_i, B_i)}$$